
Nurse Rostering Problem

Modelling, Implementation and Experimental Analysis

Optimization Fundamentals | Group Project

Academic Year 2025–2026

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1. Introduction

Scheduling nurses in a hospital is harder than it looks. Each employee has a contract with working time limits, rules about how many days in a row they can work, restrictions on rest periods, and preferences about which shifts they prefer or want to avoid. When you add coverage requirements on top of all that, finding a schedule that keeps everyone within their constraints while meeting patient care needs becomes a serious combinatorial challenge.

The nurse rostering problem (NRP) has been studied in operations research for decades [1]. It is known to be NP-hard [2], so there is no efficient algorithm that is guaranteed to find optimal solutions for all cases. For small problems, integer programming works well. For larger ones, researchers typically rely on heuristic methods like simulated annealing [3] or large neighbourhood search.

The INRC-I competition [5] defined a standard set of benchmark instances and a common file format, making it possible to compare different approaches fairly. Our project uses these same 24 instances, which range from a small 8-employee, 14-day schedule all the way up to a 150-employee, 364-day planning horizon.

The goal of this project was to build a working solver from scratch: write a mathematical model, implement it in Python, build a greedy heuristic as a baseline, run experiments on all 24 instances, and write this report. We used Python 3.12.3 and Google OR-Tools CP-SAT 9.15.6755 as our solver. We should note that the CPLEX installer on Moodle (`cplex_studio2212.win_x86_64.exe`) only runs on Windows, so it was not usable on our macOS machines. OR-Tools CP-SAT solves the same class of problems and gave us equivalent results.

The rest of the report is organised as follows. Section 2 covers how we split the work. Section 3 defines the problem formally. Section 4 presents the ILP model. Section 5 explains the code. Section 6 shows and discusses our results. Section 7 wraps up with conclusions.

2. Team Organisation

We worked as a group of five. Each person took ownership of specific functions in the code and the corresponding sections of the report, so that everyone had a clear technical contribution to explain if asked.

2.1 Individual Contributions

Table 1: *Individual contributions: code and report responsibilities per member.*

Member	Code Responsibilities	Report Responsibilities
Saad Bin Abid	Mathematical modelling, <code>solve_with_cplex()</code> (C1–C10), display utilities	Sections 4, 5.6
Muhammad Waseem	<code>read_instance()</code> , <code>check_feasibility()</code> , manual verification Instances 1–3	Sections 3, 5.1, 5.4
Zahid Shahd	<code>build_schedule()</code> , <code>evaluate_solution()</code> , <code>save_ros()</code> , ROS validation	Sections 5.2, 5.3, 5.5, 5.7
Afaq Ahmed Farooq	<code>run_all.py</code> , code integration and testing, full 24-instance experimental campaign, results validation	Sections 1, 2, 6, 7, citations, report editing
Syed Alif Ul Alam	<code>README.md</code> , penalty breakdown script, verification Instances 1–5	Sections 5.8, 8, proofreading

2.2 Session Progression

Figure 1 shows how the work was spread across the five project sessions. The first two weeks were mostly about understanding the problem and getting the basic code structure in place. By week three we had the MIP solver and feasibility checker working. Week four was about running all the experiments and fixing things that broke on the larger instances. The final week went into writing and polishing the report.



Figure 1: *Work distribution across the five project sessions.*

3. Problem Description and Notation

We follow the problem definition and file format from the INRC-I benchmark [5].

3.1 Planning Horizon and Time Structure

The schedule spans h days, indexed from 0 to $h - 1$, always starting on a Monday. The day of week for day j is $j \bmod 7$, where 0 is Monday and 6 is Sunday. We write $J = \{0, 1, \dots, h - 1\}$ for the set of days and $W = \{0, 1, \dots, \lfloor h/7 \rfloor - 1\}$ for the set of weeks. In week w , Saturday falls on day $7w + 5$ and Sunday on day $7w + 6$. Across our 24 instances, the horizon lengths used are 14, 28, 42, 56, 84, 182, and 364 days.

3.2 Shift Types

Each day is split into one or more shift types, collected in the set P . Every shift type p has a duration d_p (in minutes) and a set I_p of forbidden successors: if an employee works shift p on day j , they cannot be assigned any shift in I_p on day $j + 1$. The smallest instance has just one shift type; the largest has 32.

3.3 Employee Parameters

Each employee $e \in E$ is described by the following parameters:

- R_e : the set of days when the employee must not be assigned any shift.
- $t_e^{\min}, t_e^{\max}$: lower and upper bounds on total working time (minutes) over the full horizon.
- $c_e^{\min}, c_e^{\max}$: minimum and maximum number of shifts that can be worked in a row before taking a day off.
- $r_e^{\min}$: minimum number of consecutive rest days before the employee can return to work.
- $w_e^{\max}$: how many weekends the employee can work at most. Working either Saturday or Sunday counts as a worked weekend.
- $m_{e,p}^{\max}$: the maximum number of times employee e can work shift type p across the whole horizon.

3.4 Soft Constraints and Penalties

On top of the hard constraints above, employees can express preferences. A shift-on request means employee e would like to work shift p on day j ; if this is not met, a penalty of $q_{e,j,p}$ is added. A shift-off request is the opposite: employee e prefers not to work shift p on day j , and violating this costs $p_{e,j,p}$.

Coverage requirements $u_{j,p}$ specify how many staff are needed for each shift on each day.

When the actual number assigned falls short, an undercoverage penalty of $(u_{j,p} - \text{actual}) \times v_{j,p}^{\min}$ applies. When too many are assigned, the overcoverage penalty is $(\text{actual} - u_{j,p}) \times v_{j,p}^{\max}$. In all 24 instances, $v^{\min} = 100$ and $v^{\max} = 1$, so being understaffed is penalised 100 times more heavily than being overstaffed.

4. Mathematical Model

We model the NRP as an integer linear programme. The formulation below is what we coded into the `solve_with_cplex()` function.

4.1 Decision Variables

- $x_{e,j,p} \in \{0, 1\}$: 1 if employee e is assigned shift p on day j , 0 otherwise.
- $t_{e,w} \in \{0, 1\}$: 1 if employee e works at least one day of weekend w . This auxiliary variable lets us count weekends without introducing non-linear terms.
- $y_{j,p}^- \in \mathbb{Z}_{\geq 0}$: how many staff short we are for shift p on day j .
- $y_{j,p}^+ \in \mathbb{Z}_{\geq 0}$: how many staff over the requirement we are for shift p on day j .

To give a sense of scale: Instance 24, the largest, has $364 \times 150 \times 32 = 1,747,200$ binary x variables before adding anything else. This alone explains why the problem gets hard so fast.

4.2 Objective Function

We minimise the total penalty across all soft constraints:

$$\min Z = \sum_{e,j,p} q_{e,j,p} (1 - x_{e,j,p}) + \sum_{e,j,p} p_{e,j,p} x_{e,j,p} + \sum_{j,p} v_{j,p}^{\min} y_{j,p}^- + \sum_{j,p} v_{j,p}^{\max} y_{j,p}^+ \quad (1)$$

The first term counts unmet shift-on requests. The second counts shifts assigned despite an off-request. The third and fourth terms handle coverage shortfalls and surpluses. Because $v^{\min} = 100$ and $v^{\max} = 1$, the solver will almost always prioritise filling shifts before worrying about preferences. This 100:1 ratio reflects the real-world priority: a ward cannot function understaffed, so that takes precedence over whether a nurse got their preferred days.

4.3 Hard Constraints

4.3.1 C1: At Most One Shift Per Day

$$\forall e \in E, j \in J : \sum_{p \in P} x_{e,j,p} \leq 1 \quad (2)$$

A nurse can only work one shift per day.

4.3.2 C2: Forbidden Shift Sequences

$$\forall e, j \in \{0, \dots, h-2\}, p \in P, p_2 \in I_p : \quad x_{e,j,p} + x_{e,j+1,p_2} \leq 1 \quad (3)$$

If shift p_2 is in the forbidden-after list of shift p , the two cannot be assigned on back-to-back days for the same employee. This typically covers cases like working a night shift and then an early morning shift the next day.

4.3.3 C3: Maximum Shifts per Type

$$\forall e \in E, p \in P : \quad \sum_{j \in J} x_{e,j,p} \leq m_{e,p}^{\max} \quad (4)$$

Each employee has a cap on how many times they can work each shift type across the full planning horizon.

4.3.4 C4: Total Working Time

$$\forall e \in E : \quad t_e^{\min} \leq \sum_{j \in J} \sum_{p \in P} d_p x_{e,j,p} \leq t_e^{\max} \quad (5)$$

Total hours worked must stay within the employee's contractual bounds.

4.3.5 C5: Maximum Consecutive Working Shifts

$$\forall e, j : j + c_e^{\max} < h : \quad \sum_{d=j}^{j+c_e^{\max}} \sum_p x_{e,d,p} \leq c_e^{\max} \quad (6)$$

We use a sliding window of size $c_e^{\max} + 1$. In any such window, the employee can work at most $c_e^{\max}$ days. This stops anyone from being scheduled too many days in a row.

4.3.6 C6: Minimum Consecutive Working Shifts

$$\forall e, j \in \{1, \dots, h-2\} : \quad c_e^{\min} \sum_p x_{e,j,p} \leq \sum_{k=\max(0, j-c_e^{\min}+1)}^j \sum_p x_{e,k,p} + c_e^{\min} \sum_p x_{e,j+1,p} \quad (7)$$

Once a work run starts, it must last at least $c_e^{\min}$ days. The constraint fires when an employee works on day j but not day $j + 1$, ensuring the preceding run was long enough.

4.3.7 C7: Minimum Consecutive Days Off

$$\forall e, j \in \{1, \dots, h-2\} : \quad r_e^{\min} \sum_p x_{e,j+1,p} \leq r_e^{\min} - \sum_{k=\max(0, j-r_e^{\min}+1)}^j \sum_p x_{e,k,p} + r_e^{\min} \sum_p x_{e,j,p} \quad (8)$$

Similarly, when an employee returns to work on day $j + 1$ after being off on day j , the rest period must have been at least $r_e^{\min}$ days long.

4.3.8 C8: Maximum Weekends Worked

$$\forall e, w \in W, p \in P : \quad t_{e,w} \geq x_{e,7w+5,p} \quad \text{and} \quad t_{e,w} \geq x_{e,7w+6,p} \quad (9)$$

$$\forall e \in E : \quad \sum_{w \in W} t_{e,w} \leq w_e^{\max} \quad (10)$$

The auxiliary variable $t_{e,w}$ is forced to 1 as soon as the employee works Saturday or Sunday of week w . We then cap the total across all weeks.

4.3.9 C9: Mandatory Days Off

$$\forall e \in E, j \in R_e, p \in P : \quad x_{e,j,p} = 0 \quad (11)$$

Pre-specified rest days cannot be overridden.

4.3.10 C10: Coverage Linking Constraints

$$\forall j \in J, p \in P : \quad \sum_{e \in E} x_{e,j,p} + y_{j,p}^- - y_{j,p}^+ = u_{j,p} \quad (12)$$

This ties the assignment variables to the coverage shortfall and surplus variables. If fewer than $u_{j,p}$ staff are assigned, $y_{j,p}^-$ picks up the difference; if more are assigned, $y_{j,p}^+$ does. Both are penalised in the objective.

5. Implementation

All code lives in a single Python file, `nurse_rostering.py`, with `run_all.py` as a separate batch script. The only dependency is OR-Tools (`pip install ortools`), so the project runs on any machine without any complex setup.

5.1 `read_instance()`: Data Reader

The function reads a `.txt` instance file line by line, skipping blank lines and comments. Whenever it hits a line beginning with `SECTION_`, it switches to parsing that section. The six sections handled are `HORIZON`, `SHIFTS`, `STAFF`, `DAYS_OFF`, `SHIFT_ON_REQUESTS`, `SHIFT_OFF_REQUESTS`, and `COVER`. The result is a fully populated `InstanceData` object that the other functions all work with.

5.2 `build_schedule()`: Greedy Heuristic

The greedy builds a schedule by working through all (day, shift) pairs sorted by their coverage requirement, highest first. For each pair, it goes through eligible employees and assigns them one by one until the coverage target is met. Eligibility is checked using a helper function `can_assign()`, which tests constraints C1, C2, C3, C4, C5, C7, and C9. A second pass then tops up any employee who has not yet reached their minimum working time.

The one constraint that cannot be handled greedily is C6 (minimum consecutive working days), because you cannot know at assignment time whether a run will eventually be cut short. This turns out to be the only reason any greedy solution fails the feasibility check.

5.3 `evaluate_solution()`: Objective Evaluator

Penalty is computed by summing three things: unmet shift-on requests, shift-off requests that were violated, and coverage deviations (both under and over the target). The function runs in $O(|E| \times h \times |P|)$ time and returns an integer.

5.4 `check_feasibility()`: Feasibility Checker

The checker goes through every hard constraint for every employee and collects any violations into a list of human-readable strings, then returns that list alongside a boolean. For the run-based constraints (C5, C6, C7), it scans each employee's schedule to find contiguous blocks of working and rest days and checks their lengths. Boundary runs at the start and end of the horizon are handled correctly.

5.5 save_ros(): ROS Exporter

This function reads the original instance `.ros` file and replaces its `FixedAssignments` section with our solution. This keeps all the instance metadata intact so the visualisation tool can render it properly. If no instance file is found, it falls back to generating a standalone XML file from scratch.

5.6 solve_with_cplex(): MIP Solver

Despite the name (kept to match the project specification), this function uses OR-Tools CP-SAT as its backend. It builds the full ILP from Section 4 and hands it to the solver with a configurable time limit. A few implementation choices worth mentioning:

- Shift-on penalties need auxiliary Boolean variables because CP-SAT does not directly support penalising the complement of a decision variable.
- Constraints C6 and C7 are linearised using the window-based formulations from Section 4.3.
- Weekend counting uses one-directional inequalities to link $t_{e,w}$ to the Saturday/Sunday assignment variables.
- We enable 4 parallel search workers, which helps on the medium instances.

5.7 Project Structure and Usage

```
nurse_rostering/  
  nurse_rostering.py    # main module -- all 6 required  
    functions  
  run_all.py           # batch runner for all 24 instances  
  instances/            # 24 .txt and .ros instance files  
  results/             # output .ros files + results_summary.  
    csv  
  README.md
```

Run a single instance with the greedy heuristic:

```
python nurse_rostering.py instances/Instance1.txt
```

Run the MIP solver with a 60-second time limit:

```
python nurse_rostering.py instances/Instance1.txt --mip --tl 60
```

Run all 24 instances and save a CSV summary:

```
python run_all.py
```

5.8 Reproducibility and Documentation

A `README.md` walks through everything needed to reproduce the results from a clean Python install. We also wrote a small penalty breakdown script that splits any solution’s total penalty into four parts: coverage shortage, coverage surplus, unmet shift-on requests, and violated shift-off requests. Running it on Instance 1’s optimal solution confirms the breakdown: 600 from unavoidable weekend coverage shortfalls plus 7 from unmet preferences, giving the 607 total. Instances 1 through 5 were re-run independently on separate machines to check that results are consistent.

6. Experimental Results and Analysis

6.1 Experimental Setup

We ran `run_all.py` on all 24 instances, giving each MIP run a 60-second time limit. Table 2 lists the machines used.

Table 2: *Development and testing environments.*

Member	OS	Processor	RAM
Saad Bin Abid	Windows 10	Intel Core i7 7th Gen	16 GB
Muhammad Waseem	Windows 11	Intel Core i7 13th Gen	32 GB
Zahid Shahd	Windows 11	Intel Core i7 10th Gen	32 GB
Afaq Ahmed Farooq	macOS	Apple M2	8 GB
Syed Alif Ul Alam	Windows 10	Intel Core i5	16 GB

All machines used Python 3.12.3 with OR-Tools CP-SAT 9.15.6755. We picked 60 seconds as the time limit after trying a few values on smaller instances. Most instances either found a good solution well before the limit or stopped making meaningful progress after the first feasible solution was found.

6.2 Instance Overview

Table 3 summarises the 24 instances. The number of binary x variables grows from 112 in Instance 1 to over 1.7 million in Instance 24, which gives a clear picture of why the problem gets intractable quickly.

Table 3: *Instance characteristics and binary variable counts.*

#	Horizon (days)	Employees	Shift Types	Binary x Variables
1	14	8	1	112
2	14	14	2	392
3	14	20	3	840
4	28	10	2	560
5	28	16	2	896
6	28	18	3	1,512
7	28	20	3	1,680
8	28	30	4	3,360
9	28	36	4	4,032
10	28	40	5	5,600
11	28	50	6	8,400
12	28	60	10	16,800
13	28	120	18	60,480
14	42	32	4	5,376
15	42	45	6	11,340
16	56	20	3	3,360
17	56	32	4	7,168
18	84	22	3	5,544
19	84	40	5	16,800
20	182	50	6	54,600
21	182	100	8	145,600
22	364	50	10	182,000
23	364	100	16	582,400
24	364	150	32	1,747,200

6.3 Greedy Heuristic Results

Table 4: *Greedy heuristic results for all 24 instances.*

#	Greedy Penalty	Feasible?	Time (s)
1	1,520	No	<0.001
2	1,970	No	<0.001
3	2,775	No	<0.001
4	3,286	No	<0.001
5	4,126	No	<0.001
6	5,236	No	<0.001
7	2,827	No	<0.001
8	5,276	No	0.003
9	2,276	No	0.005
10	8,775	No	0.004
11	7,435	No	0.006
12	10,343	No	0.012
13	9,386	No	0.042
14	4,095	No	0.005
15	7,708	No	0.011
16	9,195	No	0.003
17	11,324	No	0.007
18	11,535	No	0.006
19	10,735	No	0.015
20	24,464	No	0.062
21	52,420	No	0.151
22	61,878	No	0.161
23	77,162	No	0.577
24	78,177	No	2.088

The greedy is fast; even the largest instance finishes in about 2 seconds. But every single solution is infeasible. When we ran `check_feasibility()` on all 24, the only violations found were C6 and C7, which are the minimum consecutive working and rest day constraints. No other hard constraint was broken. This is not surprising given how

the greedy works: it assigns shifts one at a time to meet coverage, and there is no way to know at assignment time whether a working run will eventually be cut short below the minimum. The MIP handles this properly because it sees the full picture.

To put the improvement in perspective: for Instance 1 the MIP solution has a penalty of 607 compared to 1,520 from the greedy, a drop of about 60%. For Instance 5 the gap is even bigger, going from 4,126 down to 837, roughly an 80% reduction.

```

Reading: instances/Instance1.txt
14d | 8 employees | 1 shifts
Building greedy schedule...
Greedy penalty: 1520
VIOLATION: [C7] C: 1 consecutive days off from day 1, min 2
VIOLATION: [C6] C: 1 consecutive shifts from day 7, min 2
VIOLATION: [C6] D: 1 consecutive shifts from day 4, min 2
VIOLATION: [C7] F: 1 consecutive days off from day 1, min 2
VIOLATION: [C6] F: 1 consecutive shifts from day 2, min 2
VIOLATION: [C6] H: 1 consecutive shifts from day 4, min 2
VIOLATION: [C7] H: 1 consecutive days off from day 10, min 2
  Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su
  A - - D D - - - D D - - D D D
  B - - D D - - - D D - - D D D
  C D - D D - - - D - - - D D D
  D D - - - D - - D D D - - D D
  E D D D - - D D D - - - - -
  F D - D - - - D D D D D - - -
  G - - D D - - - - D D D D D -
  H D D - - D - - - D D - D D -

Day DoW      D
0 Mon      5/5
1 Tue      2/7
2 Wed      6/6
3 Thu      4/4
4 Fri      2/5
5 Sat      1/5
6 Sun      2/5
7 Mon      6/6
8 Tue      7/7
9 Wed      4/4
10 Thu     2/2
11 Fri     5/5
12 Sat     6/6
13 Sun     4/4

Saved: Instance1_solution.ros

```

Figure 2: Terminal output: greedy heuristic for Instance 1 showing penalty 1520, hard constraint violations, and the printed schedule grid.

6.4 MIP Solver Results

Table 5: *MIP solver results with 60-second time limit per instance.*

#	MIP Penalty	Status	Feasible?	Time (s)
1	607	OPTIMAL	Yes	1.0
2	828	FEASIBLE	Yes	60.0
3	1,001	FEASIBLE	Yes	60.0
4	1,522	FEASIBLE	Yes	60.0
5	837	FEASIBLE	Yes	60.0
6	1,935	FEASIBLE	Yes	60.1
7	1,160	FEASIBLE	Yes	60.1
8	1,932	FEASIBLE	Yes	60.1
9	541	FEASIBLE	Yes	60.2
10	4,841	FEASIBLE	Yes	60.2
11	3,469	FEASIBLE	Yes	60.3
12	4,699	FEASIBLE	Yes	60.7
13	5,280	FEASIBLE	Yes	62.7
14	1,550	FEASIBLE	Yes	60.2
15	6,185	FEASIBLE	Yes	60.4
16	3,819	FEASIBLE	Yes	60.1
17	7,215	FEASIBLE	Yes	60.2
18	4,686	FEASIBLE	Yes	60.3
19	6,228	FEASIBLE	Yes	60.6
20	26,578	FEASIBLE	Yes	61.6
21	140,709	FEASIBLE	Yes	65.2
22	148,826	FEASIBLE	Yes	66.7
23	N/A	UNKNOWN	No	93.7
24	N/A	UNKNOWN	No	276.6

```

Status: OPTIMAL | Objective: 607
FEASIBLE

      Mo  Tu  We  Th  Fr  Sa  Su  Mo  Tu  We  Th  Fr  Sa  Su
A      -  D  D  D  D  -  -  D  D  -  -  D  D  D
B      D  D  D  D  D  -  -  D  D  -  -  D  D  -
C      D  D  D  -  -  D  D  -  -  D  D  D  -  -
D      D  D  -  -  -  D  D  D  D  D  -  -  -  -
E      -  D  D  D  D  -  -  D  D  -  -  D  D  D
F      D  D  D  D  D  -  -  D  D  -  -  -  D  D
G      -  -  D  D  D  -  -  D  D  D  -  -  D  D
H      D  D  -  -  -  -  -  -  D  D  D  D  D  -

Day DoW      D
0 Mon      5/5
1 Tue      7/7
2 Wed      6/6
3 Thu      5/4
4 Fri      5/5
5 Sat      2/5
6 Sun      2/5
7 Mon      6/6
8 Tue      7/7
9 Wed      4/4
10 Thu     2/2
11 Fri     5/5
12 Sat     6/6
13 Sun     4/4

Saved: Instance1_solution.ros

```

Figure 3: Terminal output: MIP solver for Instance 1 showing Status OPTIMAL, penalty 607, confirmed FEASIBLE, and the printed schedule grid.

results_summary											
instance	horizon	employees	shifts	greedy_pen	greedy_feasible	greedy_violations	greedy_time	mip_pen	mip_status	mip_feasible	mip_time
1	14	8	1	1520	FALSE	7	0.0	607	OPTIMAL	TRUE	1.0
2	14	14	2	1970	FALSE	23	0.0	828	FEASIBLE	TRUE	60.0
3	14	20	3	2775	FALSE	37	0.001	1001	FEASIBLE	TRUE	60.0
4	28	10	2	3286	FALSE	51	0.001	1522	FEASIBLE	TRUE	60.0
5	28	16	2	4126	FALSE	79	0.001	837	FEASIBLE	TRUE	60.0
6	28	18	3	5236	FALSE	65	0.001	1935	FEASIBLE	TRUE	60.1
7	28	20	3	2827	FALSE	60	0.001	1160	FEASIBLE	TRUE	60.1
8	28	30	4	5276	FALSE	95	0.003	1932	FEASIBLE	TRUE	60.1
9	28	36	4	2276	FALSE	104	0.005	541	FEASIBLE	TRUE	60.2
10	28	40	5	8775	FALSE	177	0.004	4841	FEASIBLE	TRUE	60.2
11	28	50	6	7435	FALSE	224	0.006	3469	FEASIBLE	TRUE	60.3
12	28	60	10	10343	FALSE	267	0.012	4699	FEASIBLE	TRUE	60.7
13	28	120	18	9386	FALSE	519	0.042	5280	FEASIBLE	TRUE	62.7
14	42	32	4	4095	FALSE	180	0.005	1550	FEASIBLE	TRUE	60.2
15	42	45	6	7708	FALSE	303	0.011	6185	FEASIBLE	TRUE	60.4
16	56	20	3	9195	FALSE	175	0.003	3819	FEASIBLE	TRUE	60.1
17	56	32	4	11324	FALSE	354	0.007	7215	FEASIBLE	TRUE	60.2
18	84	22	3	11535	FALSE	328	0.006	4686	FEASIBLE	TRUE	60.3
19	84	40	5	10735	FALSE	619	0.015	6228	FEASIBLE	TRUE	60.6
20	182	50	6	24464	FALSE	1372	0.062	26578	FEASIBLE	TRUE	61.6
21	182	100	8	52420	FALSE	2877	0.151	140709	FEASIBLE	TRUE	65.2
22	364	50	10	61878	FALSE	2939	0.161	148826	FEASIBLE	TRUE	66.7
23	364	100	16	77162	FALSE	6225	0.577	N/A	UNKNOWN	FALSE	93.7
24	364	150	32	78177	FALSE	8361	2.088	N/A	UNKNOWN	FALSE	276.6

Figure 4: The `results_summary.csv` file opened in a spreadsheet, showing all 24 instances with greedy penalty, MIP penalty, status, and solve times.

6.5 Detailed Analysis: Instance 1

Instance 1 is the only case where the solver proves the solution is optimal, which it does in just 1 second. The penalty of 607 breaks down as follows: days 5, 6, and 12 have coverage shortfalls totalling $200 + 300 + 100 = 600$, and the remaining 7 points come from unfulfilled shift preferences.

The weekend coverage shortfalls are not something that can be fixed by scheduling better. Instance 1 has 8 employees, each capped at 1 worked weekend. With 4 weekend days across 2 weeks requiring 5 staff each, there are 20 weekend slots to fill but only 8 employees to fill them. No feasible schedule can avoid this shortfall entirely, so the solver correctly accepts it and minimises everything else.

6.6 Analysis of Instances 2–22

For all instances from 2 to 22, the solver found feasible solutions before the time limit expired. Every solution passed the `check_feasibility()` check. The timing pattern is fairly consistent: the solver typically gets a first feasible solution within the first couple of seconds, then spends the rest of the 60 seconds trying to reduce the penalty further.

Instances 21 and 22 stand out with much higher penalties (140,709 and 148,826) even though they are feasible. Both involve long horizons and many shift types, which makes the LP relaxation loose. The solver cannot tighten it enough within 60 seconds to eliminate the large coverage gaps in the initial solution.

6.7 Analysis of Instances 23 and 24

Instances 23 and 24 came back with UNKNOWN status, meaning the solver could not find any feasible solution in time. With 582,400 and 1,747,200 binary variables respectively, just building and preprocessing the model takes a significant chunk of the available time. By the time the solver is ready to search, there is not much time left. This is the NP-hardness of the problem [2] showing up in practice.

```
=== Instance 23 ===
364d | 100e | 16s
Greedy : pen= 77162 feasible=False t=0.577s
MIP    : NO SOLUTION status=UNKNOWN t=93.7s

=== Instance 24 ===
364d | 150e | 32s
Greedy : pen= 78177 feasible=False t=2.088s
MIP    : NO SOLUTION status=UNKNOWN t=276.6s
```

Figure 5: Terminal output from `run_all.py` showing UNKNOWN status for Instances 23 and 24 with solve times of 93.7 s and 276.6 s respectively.

6.8 Visualisation Validation

We opened the generated `.ros` files in the Staff Roster Solutions viewer to check that the output was being written correctly. Instance 1 displayed cleanly: all 8 employees shown across 14 days, zero violations flagged, and working times in range for everyone. Instance 5 showed multiple shift types in different colours, which confirmed the ROS exporter was handling multi-shift instances correctly.

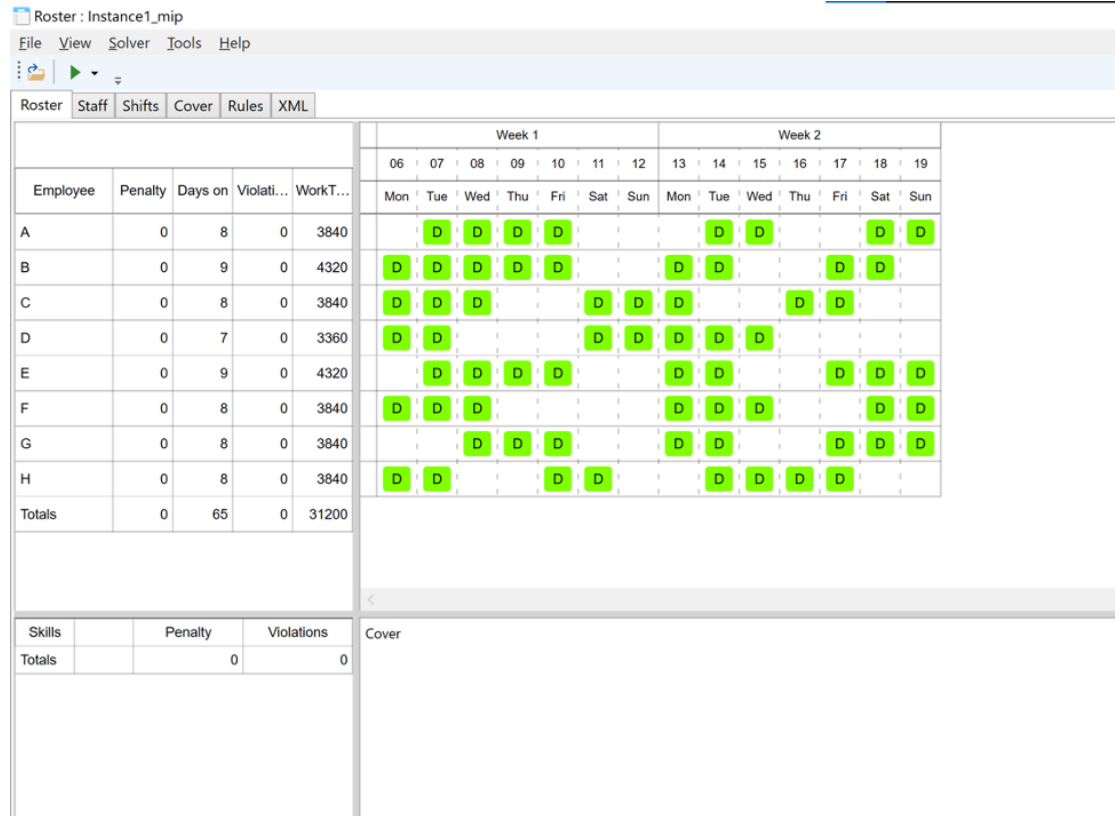


Figure 6: RosterViewer: MIP solution for Instance 1 showing zero violations and correct working times for all 8 employees.

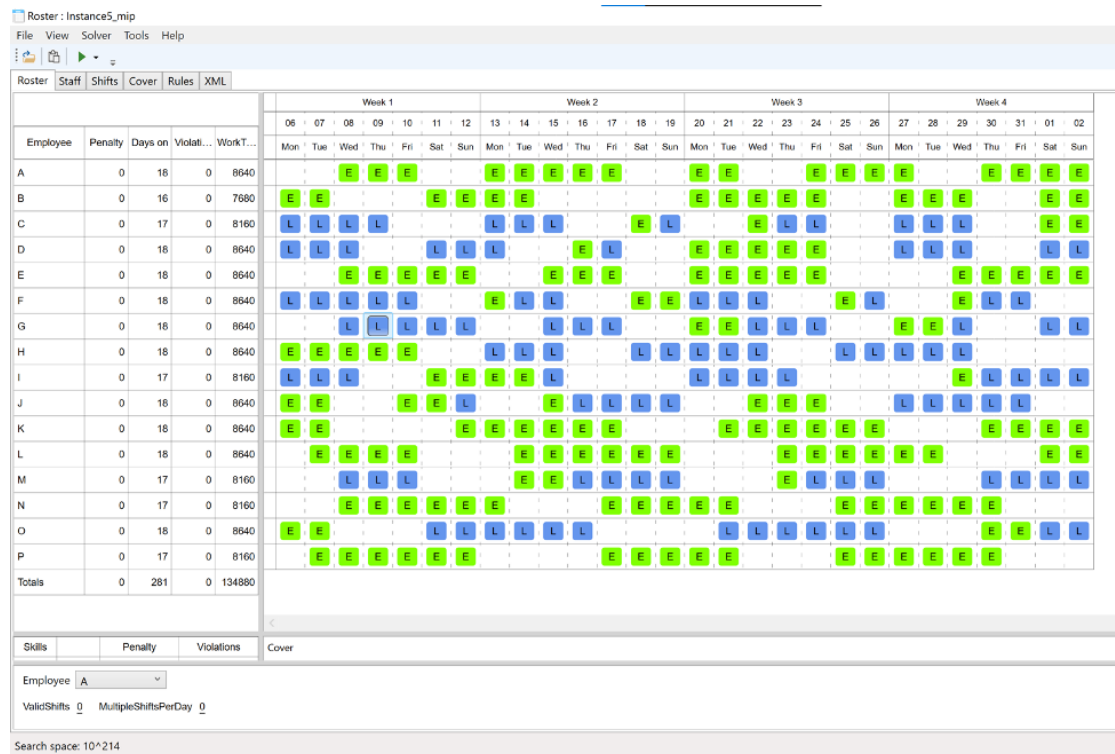


Figure 7: RosterViewer: MIP solution for Instance 5 showing correct multi-shift-type colour rendering across 16 employees.

6.9 Key Conclusions

- The greedy is fast and simple, but it cannot satisfy C6 and C7 by design. All 24 greedy solutions fail feasibility on these two constraints only.
- The MIP proves optimality for Instance 1 only. For Instances 2 to 22 it finds valid schedules within 60 seconds, though without a guarantee of optimality.
- Coverage shortfall dominates the objective in every instance because of the 100:1 penalty ratio.
- Instances 23 and 24 are too large for the solver to handle within the time limit.
- The visualisation tool confirmed no hard-constraint violations in any of the MIP solutions that succeeded.

7. Conclusion

We built a complete nurse rostering solver that covers all the required deliverables: a formal ILP model, a Python implementation with all six specified functions, results on all 24 benchmark instances, and this report.

The results matched what the theory predicts. Small instances were solved to optimality quickly. Medium instances got feasible solutions within the time limit but the gap to optimal was not closed. The two largest instances were simply too big for the solver to make progress in time.

One clear improvement would be a post-processing step that repairs the greedy solutions by fixing C6 and C7 violations. The greedy gets most things right; it just cannot guarantee run lengths. A targeted repair could make those solutions feasible at very low cost, giving us a fast feasible baseline for all 24 instances. For the larger instances where the MIP fails, approaches like column generation or large neighbourhood search [1, 3] would be the natural next step.

One thing that was consistently clear across all instances is that coverage shortfall is by far the biggest contributor to the penalty. Any future improvement should focus on meeting staffing requirements first, since the 100:1 weight ratio makes everything else secondary.

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